

Sub Group

A non-empty set H ^{of} a group G , is said to be a sub-group of G if the composition in G is also the composition of H and for this composition, H itself is a group.

Example: Find the order and sub-group generated by $(2, 7, 11)$ if group $G = \{1, 2, 4, 7, 8, 11, 13, 14\}, \times_{15}$.

Solⁿ:

Composition table,

$\times_{15}$	1	2	4	7	8	11	13	14
1	1	2	4	7	8	11	13	14
2	2	4	8	14	1	7	11	13
4	4	8	1	13	2	14	7	11
7	7	14	13	4	11	2	1	8
8	8	1	2	11	4	13	14	7
11	11	7	14	2	13	1	8	4
13	13	11	7	1	14	8	4	2
14	14	13	11	8	7	4	2	1

$$e = 1$$

For 2,

$$2^1 = 2$$

$$2^2 = 2 \times_{15} 2 = 4$$

$$2^3 = 4 \times_{15} 2 = 8$$

$$2^4 = 8 \times_{15} 2 = 1$$

~~$$2^5 = 1 \times_{15} 2 = 2$$~~

$$O(2) = 4.$$

For 7,

$$7' = 7$$

$$7^2 = 7 \times_{15} 7 = 4$$

$$7^3 = 4 \times_{15} 7 = 13$$

$$7^4 = 13 \times_{15} 7 = 1$$

$$O(7) = 4.$$

For 11,

$$11' = 11$$

$$11^2 = 11 \times_{15} 11 = 1$$

$$O(11) = 2.$$

For 2, $H_1 = \{1, 2, 4, 8\}$
Sub-group

For 7, $H_2 = \{1, 4, 7, 13\}$

For 11, $H_3 = \{1, 11\}$

For H_1 ,

$\times_{15}$	1	2	4	8
1	1	2	4	8
2	2	4	8	1
4	4	8	1	2
8	8	1	2	4

For closure property, $(a \times b) \in H_1$ i.e. 1, 2, 4 & 8 belongs to H_1 . It is algebraic structure.

For associative property, $a \times (b \times c) = (a \times b) \times c$

Let, $a = 1$, $b = 2$ & $c = 4$

$$1 \times_{15} (2 \times_{15} 4) = (1 \times_{15} 2) \times_{15} 4 \\ \Rightarrow 8 = 8.$$

It is semi-group.

For identity property, $a \times e = e \times a = a$
i.e. $a \times 1 = 1 \times a = a$

$$e = 1 \in H_1$$

It is monoid.

For inverse property, $a \times a^{-1} = a^{-1} \times a = e = 1$
& ~~$a^{-1} \in G$~~ & $a^{-1} \in H_1$.

$$1 \times_{15} 1 = 1, 1^{-1} = 1 \mid 2 \times_{15} 8 = 1, 2^{-1} = 8 \mid 4 \times_{15} 4 = 1, 4^{-1} = 4$$

$$\begin{array}{l} 1 \times \frac{1}{1} = 1 = e \\ 2 \times \frac{1}{2} = 1 = e \\ 4 \times \frac{1}{4} = 1 = e \\ 8 \times \frac{1}{8} = 1 = e \end{array}$$

$$\begin{array}{l} \cancel{2 \times 4 = 4 \times 8} \\ \Rightarrow \cancel{8 = 8} \end{array}$$

$$8 \times_{15} 2 = 1, 8^{-1} = 2 \in H_1$$

It satisfies the inverse property.

Thus, H_1 is a group and hence H_1 is sub-group of G .

For H_2 ,

X_{15}	1	4	7	13
1	1	4	7	13
4	4	1	13	7
7	7	13	4	1
13	13	7	1	4

For closure property, $(a \times b) \in H_2$ i.e. 1, 4, 7, 13 belongs to H_2 . It is algebraic structure.

For associative property, $a \times (b \times c) = (a \times b) \times c$

Let, $a = 1$, $b = 4$ & $c = 7$

$$\begin{aligned} 1 \times_{15} (4 \times_{15} 7) &= (1 \times_{15} 4) \times_{15} 7 \\ \Rightarrow 1 \times 13 &= 4 \times 7 \\ \Rightarrow 28 &= 28. \Rightarrow 13 = 13 \in H_2 \end{aligned}$$

It is semi-group.

For identity property, $a \times e = e \times a = a$
i.e. $a \times 1 = 1 \times a = a$

$$e = 1 \in H_2.$$

It is monoid.

For inverse property, $a X_{15} a^{-1} = a^{-1} X_{15} a = e = 1$
& $a^{-1} \in H_2$

$$1^{-1} = 1 \in H_2, \quad 1 X_{15} 1 = 1 = e$$

$$4^{-1} = 4 \in H_2, \quad 4 X_{15} 4 = 1 = e$$

$$7^{-1} = 13 \in H_2, \quad 7 X_{15} 13 = 1 = e$$

$$13^{-1} = 7 \in H_2, \quad 13 X_{15} 7 = 1 = e$$

It satisfies the inverse property. It is a group.

Since, H_2 is a group.

Hence, H_2 is a sub-group of G .

For H_3 ,

X_{15}	1	11
1	1	11
11	11	1

For closure property, $(a X b) \in H_3$ i.e. 1 & 11 belongs to H_3 . It is algebraic structure.

For associative property, $a X (b X c) = (a X b) X c$

$$1 X_{15} (11 X_{15} 11) = (1 X_{15} 11) X_{15} 11$$
$$\Rightarrow 1 = 1$$

It is semi-group.

For identity property, $a \times e = e \times a = a$
 $\Rightarrow a \times 1 = 1 \times a = a$

$$e = 1 \in H_3$$

It is monoid.

For inverse property, $a \times_{15} a^{-1} = e = 1$
& $a^{-1} \in H_3$

$$1^{-1} = 1 \in H_3, 1 \times_{15} 1 = 1 = e$$

$$11^{-1} = 11 \in H_3, 11 \times_{15} 11 = 1 = e$$

It satisfies inverse property. It is a group.

$\therefore H_3$ is a group.

$\therefore H_3$ is a sub-group of G .

Coset

Coset is a set obtain by multiplying or adding every element of a sub-group by a fixed element of the group.

Let G be a group and H be a sub-group of G . For any element $a \in G$, there are two types of coset:

Left Coset: The left coset of H w.r.to a is,

$$aH = \{a \cdot h \mid h \in H\}$$

$$\text{or } a \circ H = \{a \circ h \mid h \in H\}.$$

Right Coset: The right coset of H w.r.to a is,

$$Ha = \{h \cdot a \mid h \in H\}$$

$$\text{or } H \circ a = \{h \circ a \mid h \in H\}.$$

↙ Lagrange's Index Formula

Total number of distinct cosets of H in G
 $= [G : H]$

$$= \frac{|G|}{|H|}$$

$$= \frac{\text{Number of elements in } G}{\text{Number of elements in } H}$$

Q. If $G = \{1, 2, 4, 7, 8, 11, 13, 14\}$ under X_{15} and the subgroup $H_1 = \{1, 2, 4, 8\}$ & $H_2 = \{7, 11, 13, 14\}$, then find the left and right coset.

Soln:

$$G = \{1, 2, 4, 7, 8, 11, 13, 14\}$$

$$H_1 = \{1, 2, 4, 8\} \text{ \& } H_2 = \{7, 11, 13, 14\}$$

Left Coset of H_1 ,

$$1 X_{15} H_1 = 1 X_{15} \{1, 2, 4, 8\} = \{1, 2, 4, 8\}$$

$$2 X_{15} \{1, 2, 4, 8\} = \{2, 4, 8, 1\}$$

$$4 X_{15} \{1, 2, 4, 8\} = \{4, 8, 1, 2\}$$

$$7 X_{15} \{1, 2, 4, 8\} = \{7, 14, 13, 11\}$$

$$8 X_{15} \{1, 2, 4, 8\} = \{8, 1, 2, 4\}$$

$$11 X_{15} \{1, 2, 4, 8\} = \{11, 7, 14, 13\}$$

$$13 X_{15} \{1, 2, 4, 8\} = \{13, 11, 7, 14\}$$

$$14 X_{15} \{1, 2, 4, 8\} = \{14, 13, 11, 7\}$$

Here, the no. of distinct cosets is 2 and the distinct cosets are $\{1, 2, 4, 8\}$ & $\{7, 11, 13, 14\}$.

Right Coset of H_1 ,

$$H_1 X_{15} 1 = \{1, 2, 4, 8\} X_{15} 1 = \{1, 2, 4, 8\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 2 = \{2, 4, 8, 1\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 4 = \{4, 8, 1, 2\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 7 = \{7, 14, 13, 11\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 8 = \{8, 1, 2, 4\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 11 = \{11, 7, 14, 13\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 13 = \{13, 11, 7, 14\} \quad -$$

$$\{1, 2, 4, 8\} X_{15} 14 = \{14, 13, 11, 7\} \quad -$$

The distinct cosets are $\{1, 2, 4, 8\}$ & $\{7, 11, 13, 14\}$.

$\therefore$ Left Coset = Right Coset, so for H_2

$$1 X_{15} \{1, 4, 7, 13\} = \{1, 4, 7, 13\}$$

$$2 X_{15} \{1, 4, 7, 13\} = \{2, 7, 14, 11\}$$

$$4 X_{15} \{1, 4, 7, 13\} = \{4, 8, 13, 1\}$$

$$7 X_{15} \{1, 4, 7, 13\} = \{7, 13, 4, 1\}$$

$$8 X_{15} \{1, 4, 7, 13\} = \{8, 2, 11, 14\}$$

$$11 \times_5 \{1, 4, 7, 13\} = \{11, 14, 2, 8\}$$

$$13 \times_{15} \{1, 4, 7, 13\} = \{13, 7, 1, 4\}$$

$$14 \times_{15} \{1, 4, 7, 13\} = \{14, 11, 8, 2\}$$

There are two distinct cosets, they are $\{1, 4, 7, 13\}$ & $\{2, 8, 11, 14\}$.

Q. If $G = \{1, -1, i, -i\}$ w.r. to $\times$ and $H = \{1, -1\}$. Find left and ^{right} cosets?

Solⁿ:

Left Coset

Right Coset

$$1 \times \{1, -1\} = \{1, -1\}$$

$$\{1, -1\} \times 1 = \{1, -1\}$$

$$-1 \times \{1, -1\} = \{-1, 1\}$$

$$\{1, -1\} \times -1 = \{-1, 1\}$$

$$i \times \{1, -1\} = \{i, -i\}$$

$$\{1, -1\} \times i = \{i, -i\}$$

$$-i \times \{1, -1\} = \{-i, i\}$$

$$\{1, -1\} \times -i = \{-i, i\}$$

Left Coset = Right Coset.

There are two distinct cosets,

$$H_1 = \{1, -1\}$$

$$\& H_2 = \{i, -i\}.$$

Q. If group $G = \{1, 2, 3, 4, 5, 6\}$ w.r.to X_7 and $H_1 = \{1, 2, 4\}$ & $H_2 = \{1, 6\}$. Find left coset of H_1 & H_2 ?

Solⁿ:

For H_1 ,

$$1 X_7 \{1, 2, 4\} = \{1, 2, 4\}$$

$$2 X_7 \{1, 2, 4\} = \{2, 4, 1\}$$

$$3 X_7 \{1, 2, 4\} = \{3, 6, 5\}$$

$$4 X_7 \{1, 2, 4\} = \{4, 1, 2\}$$

$$5 X_7 \{1, 2, 4\} = \{5, 3, 6\}$$

$$6 X_7 \{1, 2, 4\} = \{6, 5, 3\}$$

Here, there are two left coset for H_1 i.e. $\{1, 2, 4\}$ & $\{3, 6, 5\}$

For H_2 ,

$$1 X_7 \{1, 6\} = \{1, 6\}$$

$$2 X_7 \{1, 6\} = \{2, 5\}$$

$$3 X_7 \{1, 6\} = \{3, 4\}$$

$$4 X_7 \{1, 6\} = \{4, 3\}$$

$$5 \quad X_7 \{1, 6\} = \{5, 2\}$$

$$6 \quad X_7 \{1, 6\} = \{6, 1\}$$

There are 3 distinct cosets for H_2 i.e. $\{1, 6\}$, $\{2, 5\}$ & $\{3, 4\}$.

Q. If $G = \{0, 1, 2, 3, 4, 5\}$ w.r.to $+_6$ and $H = \{0, 3\}$.
Find the coset of H ?

Solⁿ:

$$0 +_6 \{0, 3\} = \{0, 3\}$$

$$1 +_6 \{0, 3\} = \{1, 4\}$$

$$2 +_6 \{0, 3\} = \{2, 5\}$$

$$3 +_6 \{0, 3\} = \{3, 0\}$$

$$4 +_6 \{0, 3\} = \{4, 1\}$$

$$5 +_6 \{0, 3\} = \{5, 2\}$$

The distinct cosets are:

$$H_1 = \{0, 3\}$$

$$H_2 = \{1, 4\}$$

$$H_3 = \{2, 5\}$$

Q. If $G = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ and $H = \{\dots, -6, -3, 0, 3, 6, \dots\}$ be its sub-group. Find all right cosets of H in group G , $H+a$ for $a \in G$.

Solⁿ:

$$H+0 = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

$$H+1 = \{\dots, -5, -2, 1, 4, 7, \dots\}$$

$$H+2 = \{\dots, -4, -1, 2, 5, 8, \dots\}$$

$$H+3 = \{\dots, -3, 0, 3, 6, 9, \dots\}$$

Hence, the distinct cosets are

$$H, H+1 \text{ \& } H+2.$$